

EE 3203: Power Electronics
Tutorial 4: Commutation Circuits and Three Phase Rectifiers

1. For the resonant pulse commutation circuit shown in Fig. 1, assuming the initial value of voltage across the capacitor to be same as the input voltage $V_i = 230$ V,

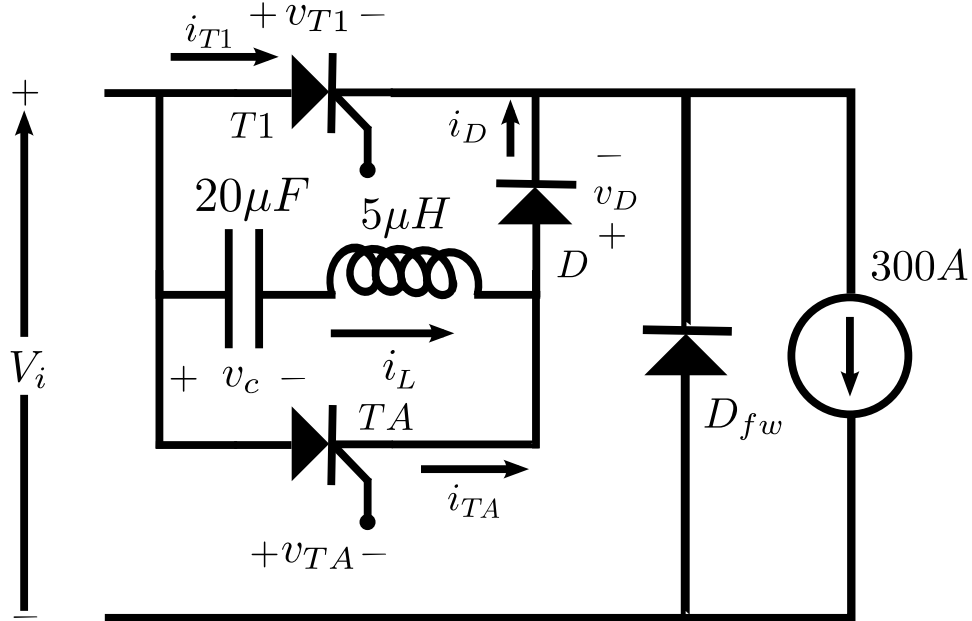


Fig. 1: Resonant pulse commutation circuit.

- a. Compute the conduction time for the auxiliary thyristor TA .
 - b. Compute the voltage v_{T1} across the main thyristor $T1$ when it gets commutated.
 - c. Compute the circuit turn-off time t_c of the main thyristor $T1$.
2. Fig. 2 shows a circuit using an impulse commutation circuit for turn-off of the main SCR $T1$. Its operation has close similarity to the resonant pulse commutation circuit. Sketch the following waveforms for one cycle of the impulse commutation circuit: gate current pulse of the main thyristor ($T1$) i_{g1} , gate current pulse of the auxiliary thyristor (TA) i_{gA} , capacitor voltage v_C , inductor current i_L , main thyristor current i_{T1} , auxiliary thyristor current i_{TA} , diode voltage v_D and freewheeling diode voltage $v_{D_{fw}}$. Assume,

- a. At time $t = 0$: $v_C(0) = V_i$, the input voltage and the freewheeling diode D_{fw} is in conduction.
- b. The main thyristor $T1$ is triggered at time $t = t_1$, $t_1 > 0$.
- c. The auxiliary thyristor TA is triggered at some time $t = t_2$ to initiate the process of commutation of main thyristor $T1$. $t_2 > t_1$ and $(t_2 - t_1) \gg \pi\sqrt{LC}$.

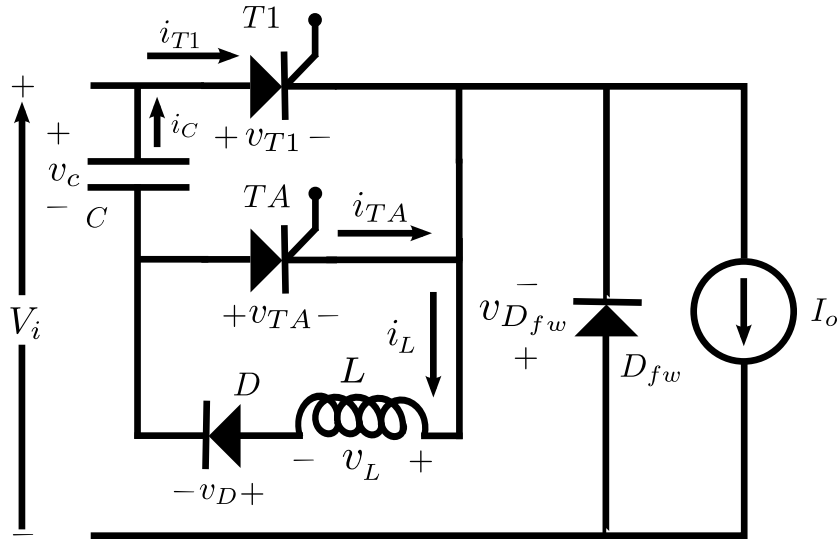


Fig. 2: Impulse commutation circuit.

- Fig. 3 shows a three pulse uncontrolled mid-point rectifier. L_p is the source inductance. The highly inductive load has been modelled as a constant current source I_o . Assume a three-phase balanced source RYB with $v_{RN} = V_m \sin(\omega t)$.

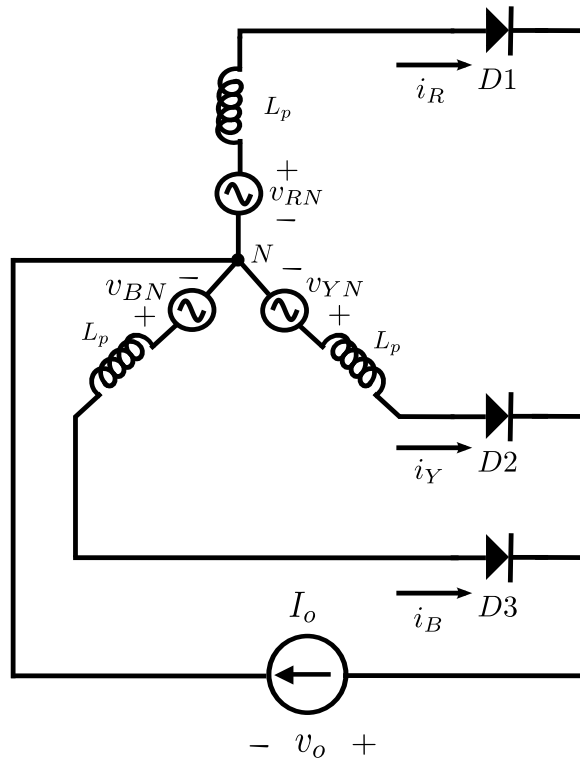


Fig. 3: Three pulse uncontrolled mid-point rectifier.

For the three pulse mid-point rectifier shown above,

- Sketch the output voltage v_o when $L_p = 0$. Indicate the devices conducting during the various sub-intervals of one fundamental cycle.
 - Compute the average value of the output voltage v_o in terms of V_m when $L_p = 0$.
 - Sketch the output voltage v_o when $L_p \neq 0$ assuming an overlap angle $u = 15^\circ$. Derive the output voltage and commutating diode's current during the overlap period of any one set of two devices. Indicate the devices conducting during the various sub-intervals of one fundamental cycle.
 - For $L_p \neq 0$, compute the average value of the output voltage v_o in terms of V_m , I_o , ω and L_p .
4. Fig. 4 shows a three-phase semi-controlled rectifier with a free-wheeling diode. The highly inductive load has been modelled as a constant current source I_o . Assume a three-phase balanced source RYB with $v_{RN} = V_m \sin(\omega t)$. Source inductance can be neglected.

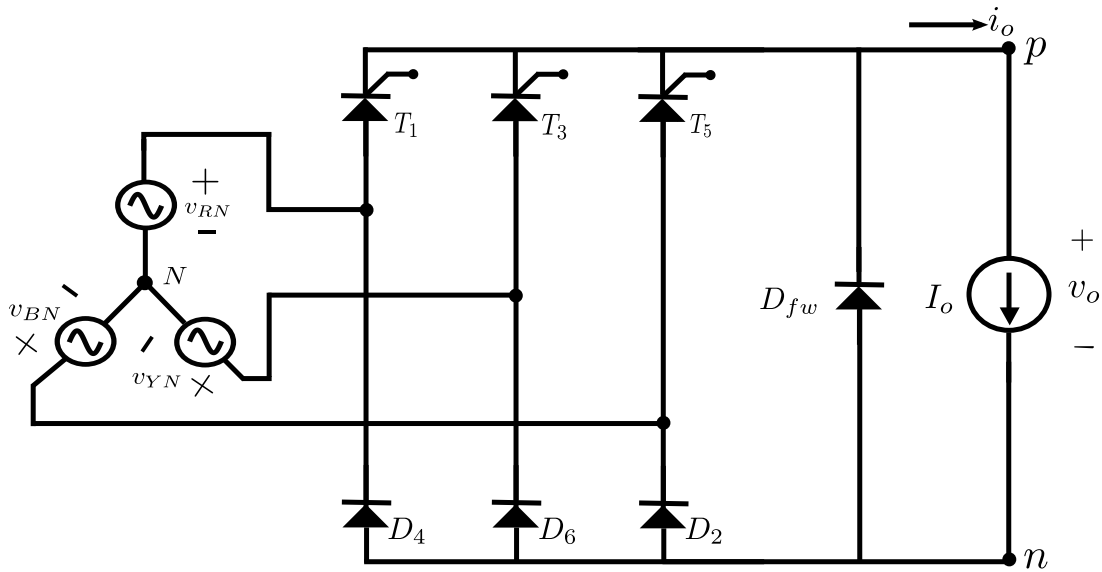


Fig. 4: Three phase semi-controlled rectifier with free-wheeling diode.

For the three-phase semi-controlled rectifier with free-wheeling diode shown above,

- Identify the valid range of firing angle α .
- Sketch the output voltage for firing angles $\alpha = 30^\circ$ and $\alpha = 90^\circ$. Indicate the conducting devices in each case. What will happen if the free-wheeling diode D_{fw} is not present?
- Derive the expression for the average value of the output voltage v_o in terms of V_m and α .